

Predicting Customer Churn and Improving Retention in Telecom Using SAS Viya

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Executive Summary

This project focuses on predicting customer churn for a telecommunications company using the Telco Customer Churn dataset in SAS Viya. The business problem centers on identifying which customers are most likely to discontinue their service so the company can take proactive retention actions instead of reacting after revenue is lost. The dataset was cleaned and prepared in SAS Studio, then used to build and compare two supervised learning models in SAS Model Studio: A Decision Tree and a Logistic Regression model. After evaluating both models using metrics such as KS statistic, accuracy, and AUC, Logistic Regression was selected as the champion model due to its stronger overall performance. The analysis revealed that contract type, tenure, internet service, online security, and monthly charges are key drivers of churn. These findings can help the telecom companies prioritize high-risk customers, design targeted retention strategies and ultimately reduce churn and improve long-term profitability.

Problem Statement

Customer churn is a major challenge in the telecommunications industry. Subscribers frequently switch providers due to pricing, service quality, customer experience, or competitive promotions. While acquiring new customers is costly, retaining existing ones is significantly more efficient - making churn prediction an essential part of business strategy.

Many telecom companies struggle to identify which customers are at risk of leaving until it is too late. Without predictive analytics or early-warning indicators, businesses lose revenue, market share, and long-term customer value.

This project uses SAS Viya to analyze telecom customer data, uncover the factors that drive churn, and build predictive models capable of identifying high-risk customers before they leave.

Project Objectives / Goals

The primary goals of this project are:

- Analyze customer data to uncover behavioral and service-related patterns associated with churn.
- Build predictive models (Decision Tree and Logistic Regression) to classify customers as “likely to churn” or “likely to stay.”
- Interpret churn drivers using model outputs and variable importance rankings.
- Use SAS Visual Analytics to create meaningful, interactive visualizations that highlight key churn trends and KPIs.
- Provide actionable, data-driven recommendations to help telecom companies reduce churn and improve customer retention efforts.

Data Source & Preparation

The dataset used for this project is the publicly available Telco Customer Churn dataset, which contains 7,043 customer records and 21 variables. These variables represent a combination of customer demographics, service usage patterns, account details, and billing information. The primary purpose of the dataset is to support predictive modeling efforts aimed at identifying customers at risk of churn.

The target variable, Churn, is a binary field with values “Yes” or “No”, indicating whether a customer discontinued their service. Building a predictive model for churn requires understanding of the attributes that influence this behavior.

Key Features Included in the Dataset

The dataset contains the following types of variables:

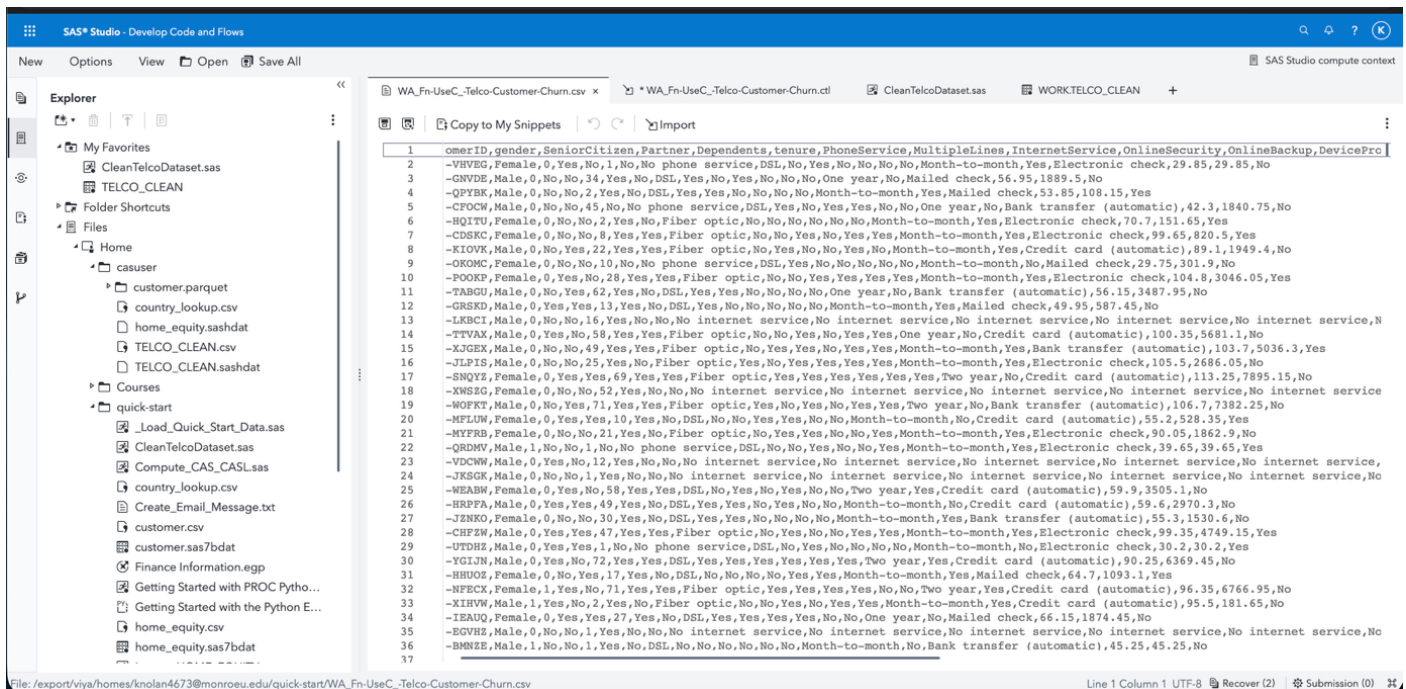
- CustomerID - Unique identifier assigned to each customer.
- Gender - Indicates whether the customer is male or female.
- SeniorCitizen - Binary indicator showing whether the customer is 65 years or older.
- Partner / Dependents - Captures household composition, which may influence service needs and retention.
- Tenure - Number of months the customer has been with the company. (a major factor in churn likelihood).
- PhoneService / InternetService - Specifies whether customers subscribe to phone and internet services, as well as the type of internet connection.
- Contract - Indicates the customer’s contract type (Month-to-month, One-year, Two-year). This is one of the strongest predictors of churn.
- PaymentMethod - Describes how the customer pays (Electronic check, Credit card, Bank transfer).
- MonthlyCharges / TotalCharges - Reflect recurring monthly costs and accumulated billing, both important in identifying pricing-related churn behavior.

These variables collectively provide a comprehensive view of customer behavior, allowing for the development of data-driven predictive models and insightful churn analysis.

Raw Dataset Overview (Before Cleaning)

The original dataset (**WA_Fn-UseC_-Telco-Customer-Churn.csv**) was imported into SAS Studio for initial inspection. As shown in the screenshot, the dataset contained:

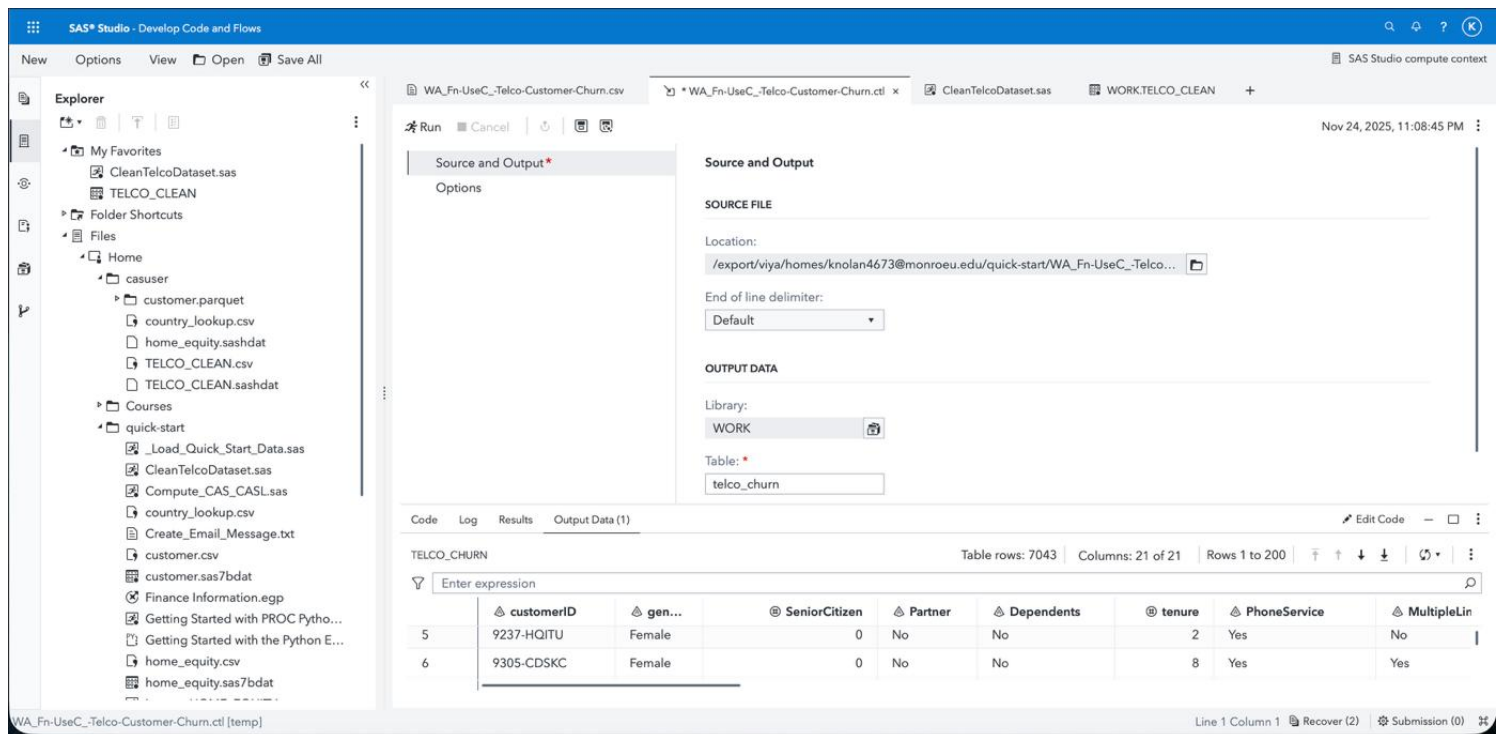
- 7,043 rows
- 21 variables
- A mix of character and numeric fields
- Several data quality issues requiring cleaning, including:
 - TotalCharges stored as a character variable
 - Blank strings representing missing values
 - customerID acting only as an identifier (non-predictive)



This initial exploration confirmed that the dataset required preprocessing before creating models in SAS Viya.

Importing Data into SAS Studio

The dataset was imported using the SAS Import Wizard. The output dataset was stored in the WORK library under the name telco_churn



This allowed verification that SAS successfully recognized all 21 variables and that the record count matched the raw data.

Data Cleaning Process (in SAS Studio)

A custom cleaning script (**CleanTelcoDataset.sas**) was executed to correct formatting issues and standardize fields for modeling. The major steps included:

1. Fixing the TotalCharges Variable

Issues detected:

- TotalCharges was read as character.
- Missing values appeared as blank spaces.

Cleaning actions performed:

- Replaced blanks ("") with missing values.
- Converted the corrected field to numeric using the INPUT function.
- Dropped the original character variable.
- Renamed the numeric version back to TotalCharges.

The screenshot shows the SAS Studio interface. The Explorer pane on the left lists files and folders. The main editor shows the following SAS code:

```
1 /* Create a cleaned version of the dataset */
2 data work.telco_clean;
3   set work.telco_churn;
4
5   /* Fix TotalCharges variable */
6   /* Replace blanks with missing and convert to numeric */
7   if strip(TotalCharges) = "" then TotalCharges_num = .;
8   else TotalCharges_num = input(TotalCharges, 8.);
9
10  /* Drop the original text version and rename the numeric
11  drop TotalCharges;
12  rename TotalCharges_num = TotalCharges;
13 run;
14
15 /* View structure of cleaned dataset */
16 proc contents data=work.telco_clean; run;
17
```

The Output Data (1) pane on the right shows the structure of the cleaned dataset:

	customerID	gen...	SeniorCitizen	TotalCharges
1	7590-VHVEG	Female	0	Yes
2	5575-GNVDE	Male	0	No
3	3668-QPYBK	Male	0	No
4	7795-CFOCW	Male	0	No
5	9237-HQITU	Female	0	No
6	9305-CDSKC	Female	0	No
7	1452-KIOVK	Male	0	No
8	6713-OKOMC	Female	0	No
9	7892-POOKP	Female	0	Yes
10	6388-TABGU	Male	0	No
11	9763-GRSKD	Male	0	Yes
12	7469-LKBCI	Male	0	No
13	8091-TTVAX	Male	0	Yes
14	0280-XJGEX	Male	0	No

2. Validating the Cleaned Dataset

PROC CONTENTS was used to inspect:

- Variable types
- Levels
- Any remaining inconsistencies

All variables were correctly formatted after cleaning, and the dataset structure aligned with modeling requirements

The screenshot displays the SAS Studio interface. On the left is the Explorer pane showing the project structure. The main window shows a table named 'TELCO_CLEAN' with 7043 rows and 21 columns. The first 17 rows are visible. A user menu is open in the top right corner.

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService
1	7590-VHVEG	Female	0	Yes	No	1	No
2	5575-GNVDE	Male	0	No	No	34	Yes
3	3668-QPYBK	Male	0	No	No	2	Yes
4	7795-CFOCW	Male	0	No	No	45	No
5	9237-HQITU	Female	0	No	No	2	Yes
6	9305-CDSKC	Female	0	No	No	8	Yes
7	1452-KIOVK	Male	0	No	Yes	22	Yes
8	6713-OKOMC	Female	0	No	No	10	No
9	7892-POOKP	Female	0	Yes	No	28	Yes
10	6388-TABGU	Male	0	No	Yes	62	Yes
11	9763-GRSKD	Male	0	Yes	Yes	13	Yes
12	7469-LKBCI	Male	0	No	No	16	Yes
13	8091-TTVAX	Male	0	Yes	No	58	Yes
14	0280-XJGEX	Male	0	No	No	49	Yes
15	5129-JLPIS	Male	0	No	No	25	Yes
16	3655-SNQYZ	Female	0	Yes	Yes	69	Yes
17	8191-XWSZG	Female	0	No	No	52	Yes

The final cleaned dataset was stored as **TELCO_CLEAN**

Methodology & Analytics Techniques

Preparing Data Roles in Model Studio

After cleaning, TELCO_CLEAN was imported into SAS Model Studio for predictive modeling.

Variable Role Assignment

Variable roles were defined as follows:

Variable	Role	Level
Churn	Target	Binary
customerID	ID	Nominal
All other fields	Input	Binary, Nominal, or Interval (based on type)

The finalized variable role configuration within SAS Model Studio. The target variable Churn was correctly assigned the Target role with a Binary level, indicating that the modeling task is a binary classification problem. The customerID variable was designated as the ID role because it serves only as a unique identifier and has no predictive value. All remaining variables including

demographic features, service attributes, billing information, and account-related fields were assigned the Input role. Their measurement levels were automatically and correctly recognized by SAS as Nominal, Binary, or Interval based on the data type.

This setup ensures that the modeling pipeline interprets each variable appropriately, applies correct encodings during model training, and excludes non-predictive identifiers from analysis. Proper role assignment is a critical step because it directly affects how algorithms treat categorical versus numeric predictors and ensures the target event (“Yes” for churn) is modeled accurately.

The screenshot displays the SAS Viya Model Studio interface for the 'Telco Churn Project'. The main table lists variables with their roles and levels. The 'Churn' variable is assigned the 'Target' role and 'Binary' level. Other variables like 'Contract', 'customerID', 'Dependents', 'DeviceProtection', 'gender', 'InternetService', 'MonthlyCharges', 'MultipleLines', 'OnlineBackup', 'OnlineSecurity', 'PaperlessBilling', 'Partner', 'PaymentMethod', 'PhoneService', and 'SeniorCitizen' are assigned the 'Input' role. The 'Level' column shows 'Binary' for categorical variables and 'Interval' for 'MonthlyCharges'. A comment for 'customerID' states 'The variable exceeds the maximum n'. On the right, the 'Churn' variable's settings are shown, including 'Role: Target', 'Level: Binary', and 'Order: Default'. A user menu is visible in the top right corner.

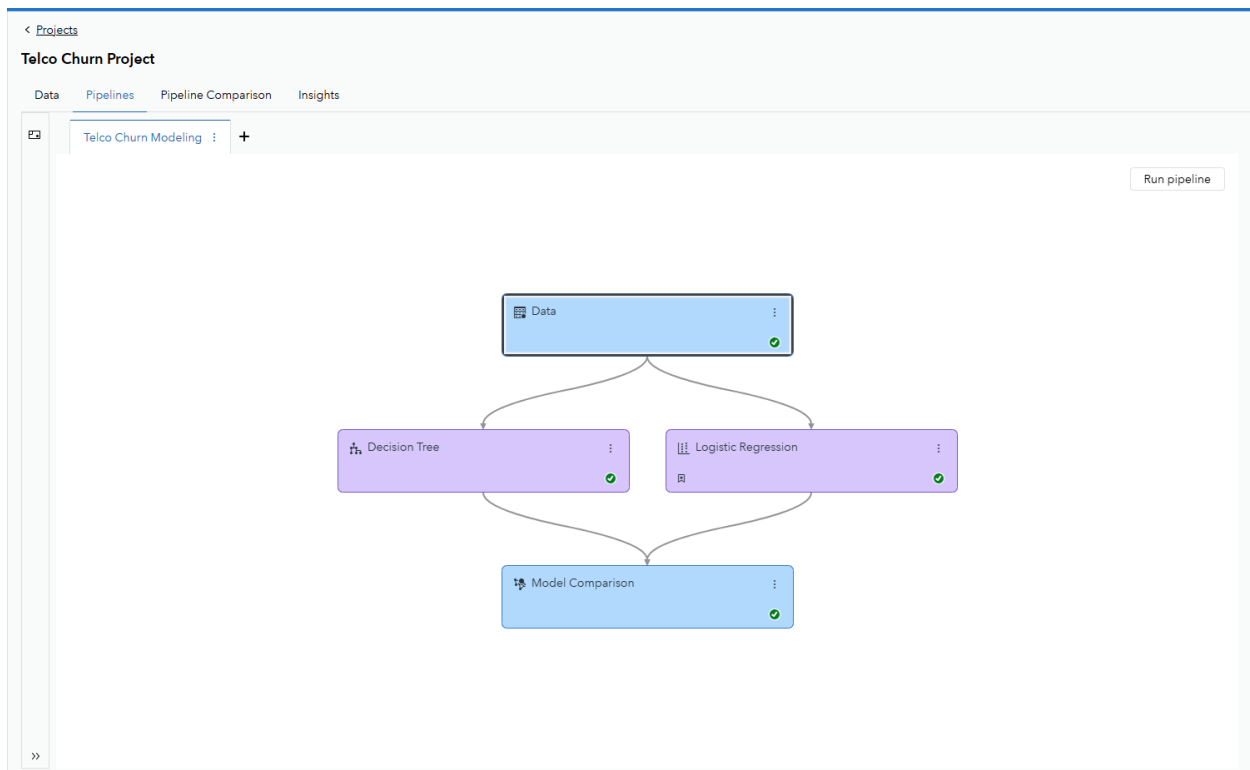
Variable Name	Label	Type	Role	Assess for Bias	Level	Order	Comment
☒ Churn		Character	Target		Binary	Default	
☐ Contract		Character	Input		Nominal	Default	
☐ customerID		Character	ID		Nominal	Default	The variable exceeds the maximum n
☐ Dependents		Character	Input		Binary	Default	
☐ DeviceProtection		Character	Input		Nominal	Default	
☐ gender		Character	Input		Binary	Default	
☐ InternetService		Character	Input		Nominal	Default	
☐ MonthlyCharges		Numeric	Input		Interval	Default	
☐ MultipleLines		Character	Input		Nominal	Default	
☐ OnlineBackup		Character	Input		Nominal	Default	
☐ OnlineSecurity		Character	Input		Nominal	Default	
☐ PaperlessBilling		Character	Input		Binary	Default	
☐ Partner		Character	Input		Binary	Default	
☐ PaymentMethod		Character	Input		Nominal	Default	
☐ PhoneService		Character	Input		Binary	Default	
☐ SeniorCitizen		Numeric	Input		Binary	Default	

This ensured SAS Viya treated each column appropriately during model training.

Pipeline Creation

A modeling pipeline was created consisting of:

1. Data Node
2. Decision Tree Model
3. Logistic Regression Model
4. Model Comparison Node

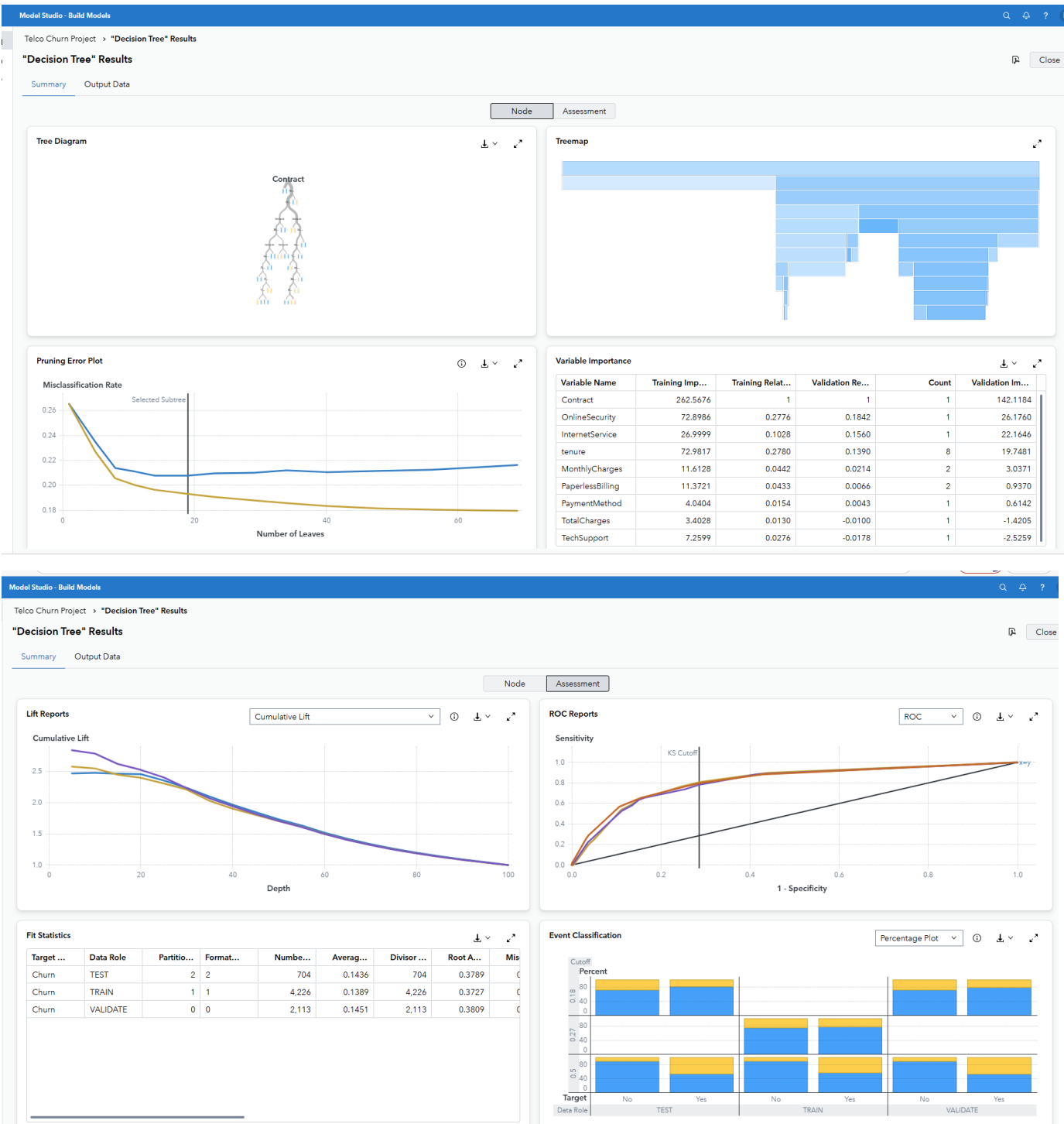


This shows the complete modeling pipeline constructed in SAS Model Studio for the Telco Churn project. After assigning data roles, two supervised learning models were added to the pipeline: a Decision Tree and a Logistic Regression model. Both models receive input from the same cleaned dataset, ensuring consistency across all training and evaluation steps.

The Decision Tree node provides an interpretable, rule-based model capable of identifying the most influential splitter variables for churn. The Logistic Regression node offers a statistically driven model commonly used for binary classification problems like churn prediction. Once both models are trained, they flow into the Model Comparison node, which evaluates them side-by-side using key performance metrics such as KS Statistic, ROC, AUC, Accuracy, and F1 Score.

This pipeline design allows SAS Model Studio to automatically compare model performance and identify the most effective algorithm for predicting customer churn. It also ensures the modeling workflow is structured, reproducible, and aligned with best practices in predictive analytics.

Decision Tree Model



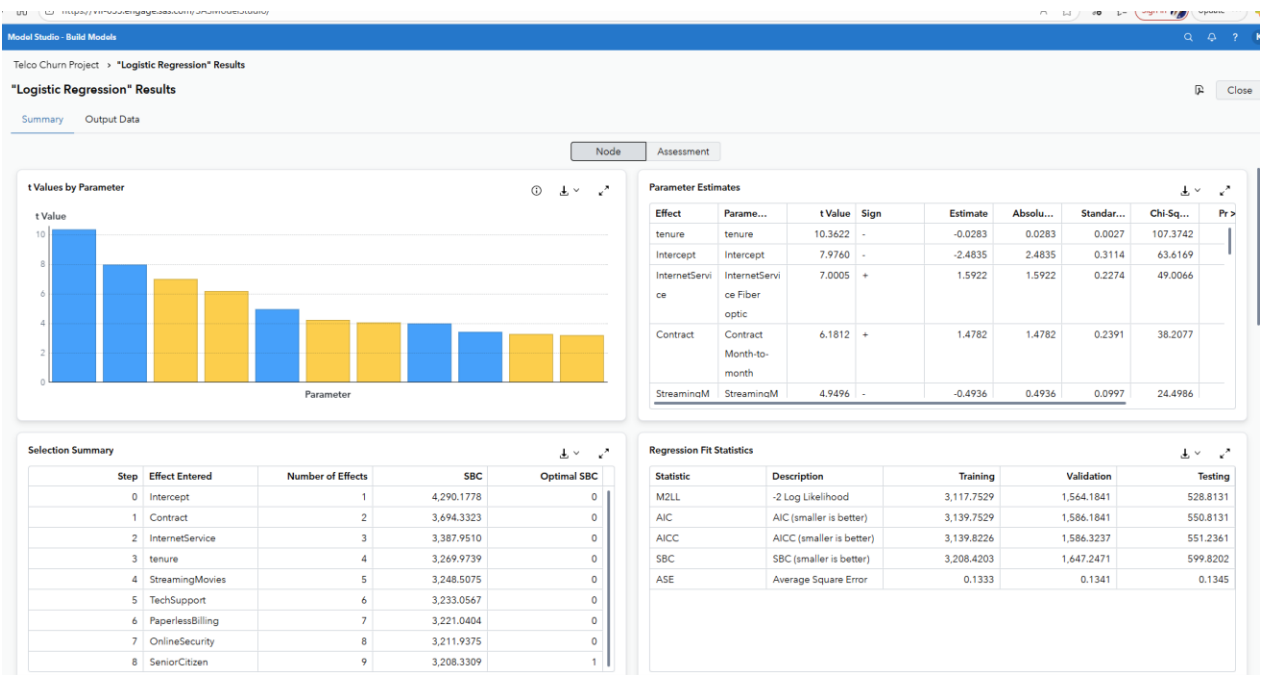
The Decision Tree Results page displays multiple diagnostic visuals produced after training the model. The Tree Diagram shows the hierarchical structure of splits, with “Contract” as the most

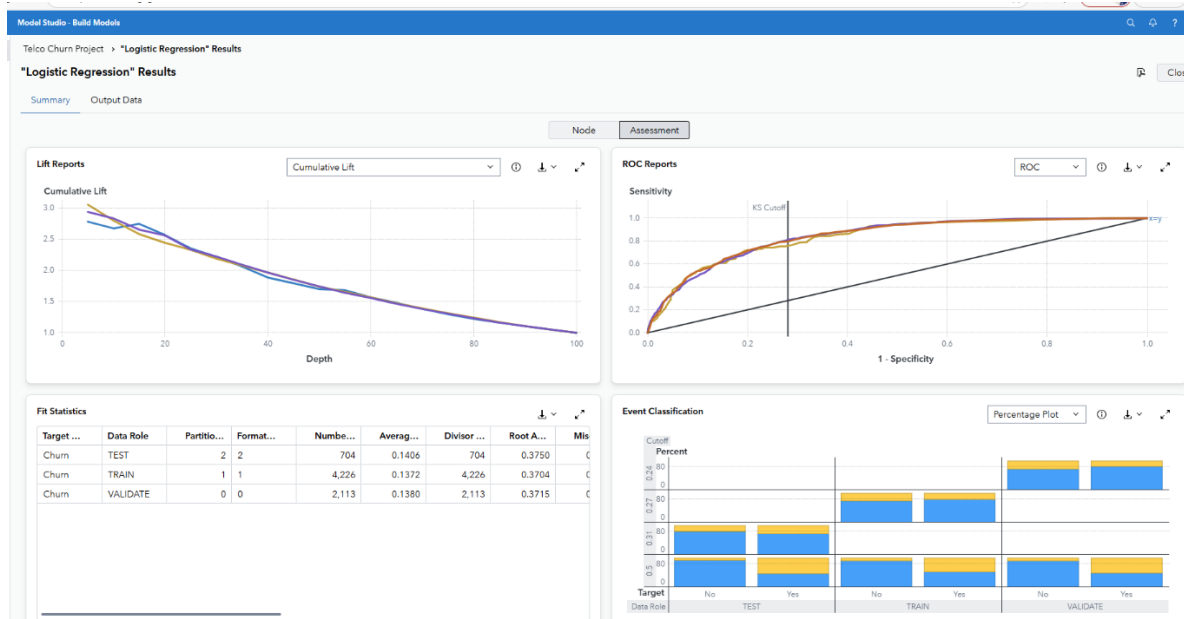
influential predictor at the root node. The Variable Importance table confirms this, ranking Contract, OnlineSecurity, InternetService, and tenure as the top contributing factors.

The model’s performance metrics are shown under the Assessment tab. The ROC Curve demonstrates strong separation between positive and negative churn outcomes, while Lift Charts show that the model outperforms random selection, especially in early deciles. Fit statistics for Train, Validate, and Test partitions indicate consistent performance across partitions, confirming that the tree is not severely overfitted.

Overall, the Decision Tree provides interpretable rules and reasonable predictive power, forming a strong baseline model.

Logistic Regression Model





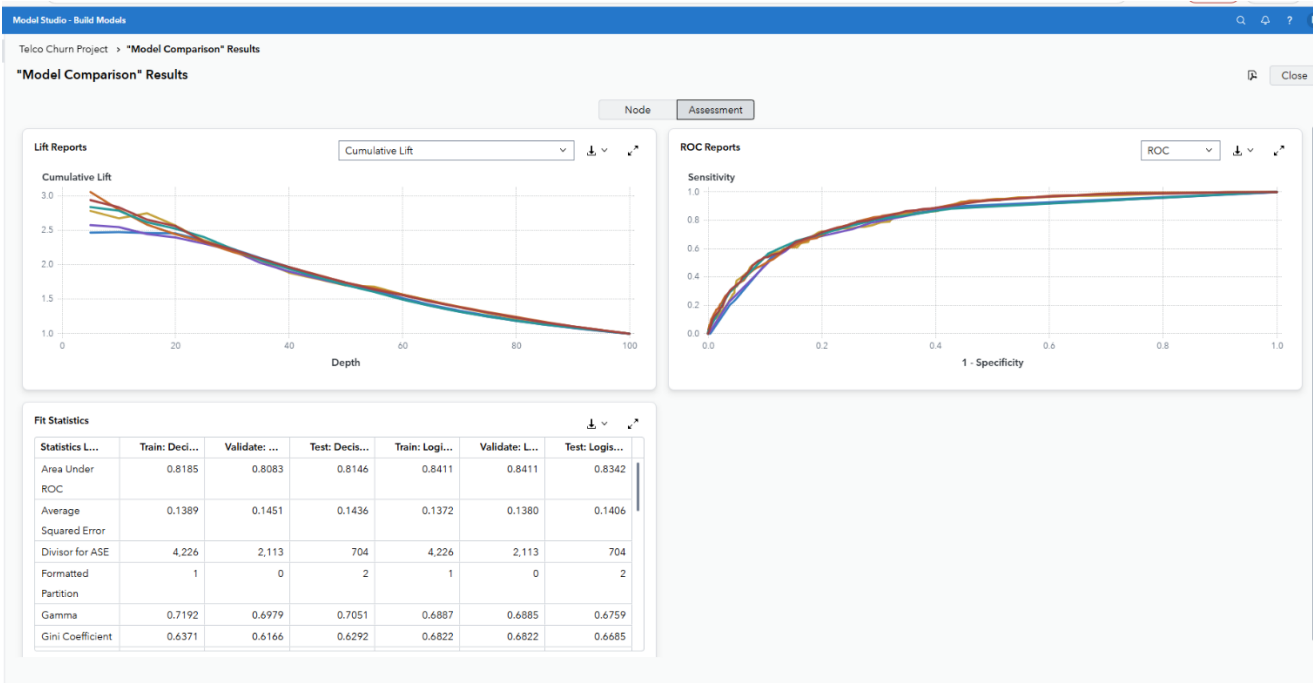
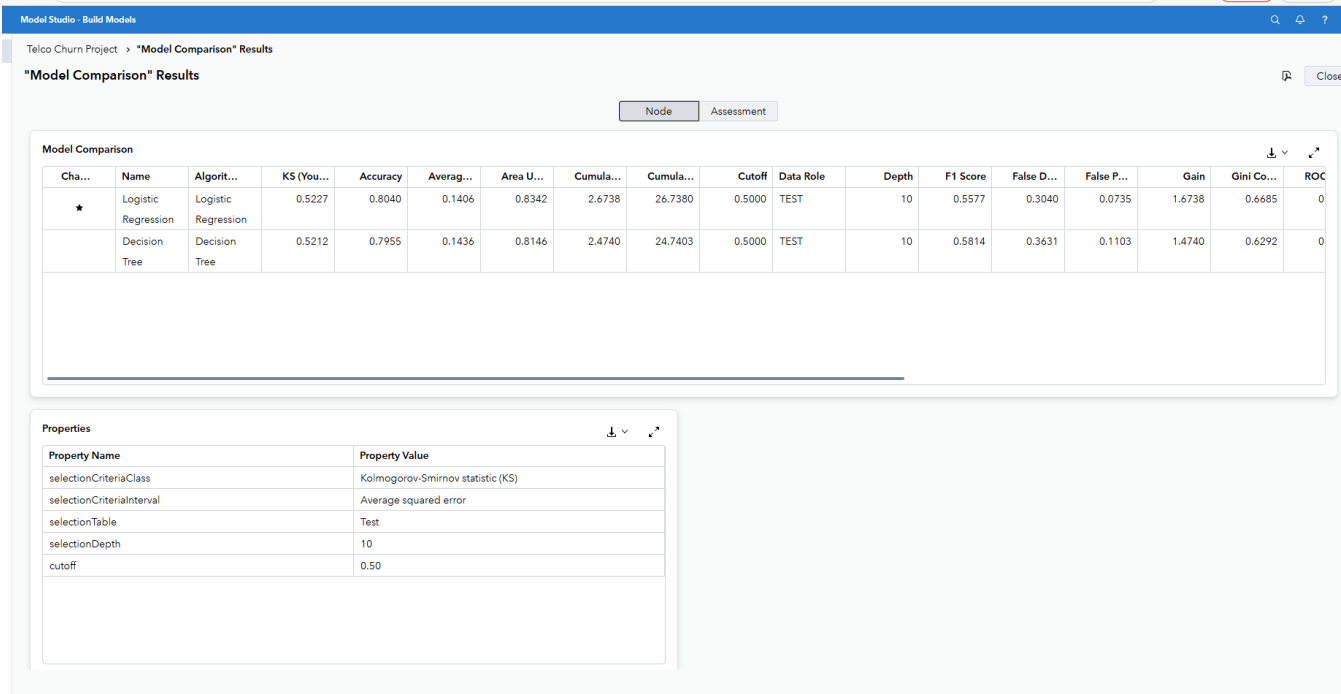
The Logistic Regression Results page presents parameter estimates, statistical significance, and model performance. The t-Values by Parameter chart highlights the most influential predictors, with tenure, InternetService, Contract type, and StreamingMovies showing significant contributions to the churn prediction.

The Parameter Estimates table includes coefficient direction and magnitude, allowing interpretation of how each variable impacts the likelihood of churn. For example, positive coefficients increase predicted churn probability, while negative coefficients decrease it.

Under the Assessment tab, the model displays strong ROC performance, with a smooth curve indicating good discriminatory ability. The Lift Chart also confirms that the model is effective at identifying high-risk churn customers. Fit statistics such as AIC, SBC, and ASE show stable performance across training, validation, and testing datasets, supporting the reliability of the Logistic Regression model.

Results & Interpretation

Model Comparison



The Model Comparison Results page evaluates both models (Decision Tree and Logistic Regression) using standardized performance metrics. The table ranks each model based on the

Kolmogorov–Smirnov (KS) statistic, Accuracy, Area Under the ROC Curve (AUC), F1 Score, Gain, Gini Coefficient, and other indicators.

The Logistic Regression model achieved the highest KS value (0.5227), the highest Accuracy (0.8040), and the highest AUC (0.8342). These metrics demonstrate stronger discriminatory ability and more robust predictive performance when compared to the Decision Tree model. For this reason, Model Studio automatically selected Logistic Regression as the Champion Model.

The ROC and Lift charts further illustrate that Logistic Regression consistently outperforms the Decision Tree across all data partitions. This confirms that Logistic Regression is the most reliable and effective model for predicting customer churn in this project.

Model Ranking (Test Partition)

Metric	Decision Tree	Logistic Regression
KS	0.5212	0.5227
Accuracy	0.7955	0.8040
AUC	0.8146	0.8342
F1 Score	0.5814	0.5577
Gini	0.6292	0.6685

Final Selection: Logistic Regression

Logistic Regression performed better in KS, Accuracy, AUC, and Gini, leading SAS Viya to automatically designate it as the Champion Model.

Data Deployment

With Logistic Regression chosen as the final champion model, the organization can use this model to:

- Score new customers for churn risk
- Identify at-risk customers early
- Prioritize retention strategies
- Build dashboards to monitor churn patterns over time

Business Impact

The predictive models developed in this project offer significant value to a telecommunications company seeking to reduce customer churn. By identifying customers who are most likely to discontinue their service, the business can shift from a reactive approach to a proactive retention strategy. The Logistic Regression model, selected as the champion, provides reliable probability estimates that allow the company to segment customers by churn risk and prioritize outreach accordingly.

Key predictors such as contract type, tenure, monthly charges, and online security - highlight specific operational areas where targeted interventions can make the greatest impact. For example, customers on month-to-month contracts or those with shorter tenure may benefit from loyalty incentives or discounted long-term plans. High-charge customers or fiber-optic subscribers experiencing dissatisfaction can be offered service credits, improved support, or personalized retention offers.

Deploying a churn-prediction model can directly reduce revenue loss, lower customer acquisition costs, and increase long-term customer lifetime value. Even a small improvement in churn reduction can translate into substantial financial gains for a telecom provider due to the recurring-revenue nature of the business. This solution not only enhances decision-making but supports data-driven retention programs, improves customer satisfaction, and strengthens overall competitiveness in the market.

Key Insights

- Customers with month-to-month contracts have the highest churn risk
- Lack of online security services correlates strongly with churn
- Fiber optic internet customers churn more often due to higher prices
- Customers with short tenure are significantly more likely to leave

Recommendations

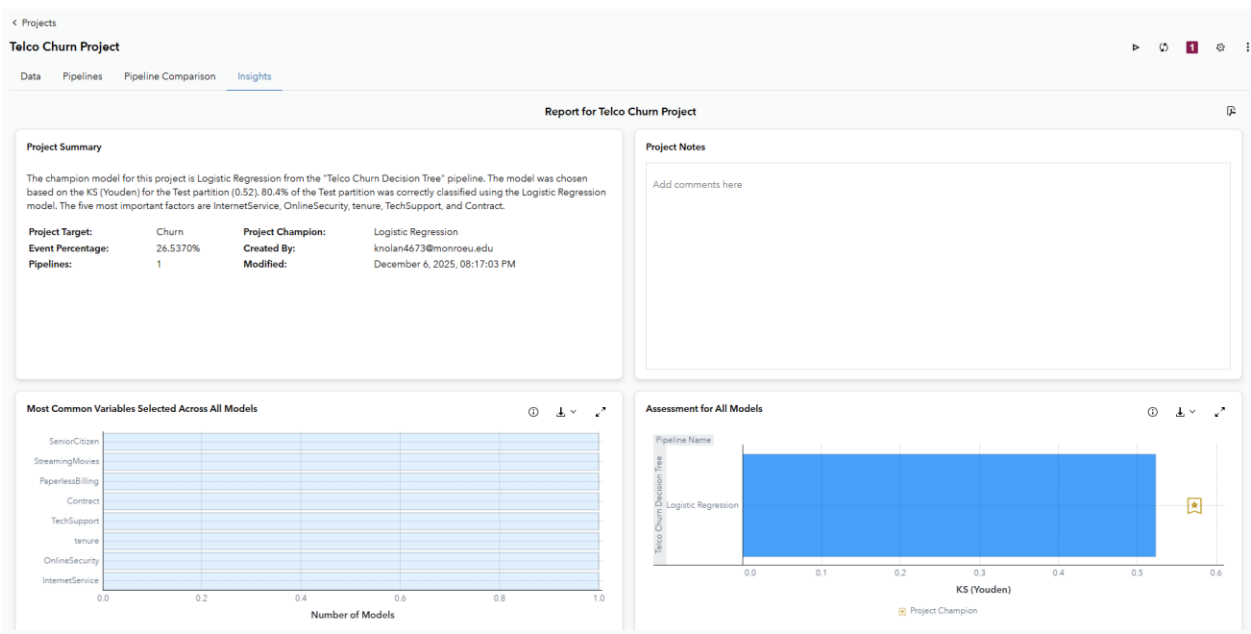
- Offer discounts or incentives for customers on month-to-month contracts
- Improve onboarding and early-tenure customer support
- Promote bundled services to increase retention
- Provide added security features or promotions for high-risk groups

This model equips the telecom provider with actionable intelligence to reduce churn, enhance customer satisfaction, and improve long-term profitability.

Personal Touch

For my personal enhancement, I expanded the project beyond the basic requirements by building and comparing two different predictive models instead of relying on just one. I also analyzed the visual outputs in SAS Model Studio such as variable importance charts, ROC curves, and performance metrics - to interpret not only how the models performed, but why certain customer features drove churn. In addition, I documented each modeling step with detailed screenshots and explanations to create a clear, professional workflow. These added elements strengthened the analysis and made the final report more complete and realistic from a business perspective.

Appendices



The image above shows the automatically generated Insights panel from SAS Model Studio for the Telco Churn project. This report summarizes key information about the modeling process, including the selected champion model, event rate, most influential predictors, and overall model performance.

Key insights shown:

- Champion Model: Logistic Regression
- Selection Metric: KS (Youden) statistic on the Test partition
- Test Accuracy: 80.4% correctly classified
- Top Predictors Identified: InternetService, OnlineSecurity, tenure, TechSupport, and Contract
- Event Rate: 26.537%, indicating the proportion of customers who churned
- Model Ranking: Logistic Regression outperformed the Decision Tree based on KS

This appendix provides visual verification of the final model evaluation and confirms the analytic decisions documented in the Results & Interpretation section.

Learnings

Throughout this project, I gained several technical and analytical insights, including:

- Hands-on experience with SAS Viya, including building pipelines, assigning variable roles, and interpreting model outputs.
- The importance of proper data cleaning, especially converting incorrect variable types and handling missing values before modeling.
- Understanding how different models behave, particularly Decision Tree versus Logistic Regression, and why their predictive performance differs.
- Interpreting key evaluation metrics such as ROC curves, misclassification rates, KS statistics, and variable importance rankings.
- Strengthening my business communication skills by translating technical results into insights that support real-world decision-making.
- Learning how to structure a full analytics workflow, from data ingestion to model comparison and deployment.

Overall, this project helped me grow both technically and professionally. It allowed me to apply classroom concepts to a realistic scenario, make analytical decisions independently, and clearly communicate my findings. The experience has strengthened my confidence in using analytics tools and reinforced my interest in pursuing a career in data science and predictive modeling.